ExpNo 7- Graph Traversal

GRAPH.C

```c
#include <stdio.h>
#include <conio.h>

#define MAX 10

int graph[MAX][MAX];
int visited[MAX];
int n;

void DFS(int vertex)
{
    int i;

    visited[vertex] = 1;
    printf("%d ", vertex);

    for (i = 0; i < n; i++)
    {
        if (graph[vertex][i] == 1 && visited[i] == 0)
        {
            DFS(i);
        }
    }
}

void main()
{
    int i, j, start;

    clrscr();

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("\nEnter adjacency matrix:\n");

    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
        {
            scanf("%d", &graph[i][j]);
        }
```

GRAPH.C

```
    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
        {
            scanf("%d", &graph[i][j]);
        }
    }

    for (i = 0; i < n; i++)
    {
        visited[i] = 0;
    }

    printf("\nEnter starting vertex (0 to %d): ", n - 1);
    scanf("%d", &start);

    printf("\nDFS Traversal: ");
    DFS(start);

    getch();
}
```

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